

Deployment Profiles: Evaluated Configurations from Part 2

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In Part 2, we evaluated specific configurations of the Cognitive Integrity Framework to understand how different tuning parameters affected security and performance outcomes. The following profiles are derived directly from the **Parameter Sensitivity Analysis** (Part 2, Section 5.3) and **Architecture-Specific Results** (Part 2, Section 5.1).

Profile A: The “Internal Tool” Baseline (Low Latency) I

This profile corresponds to the “High Usability” configuration tested in the sensitivity analysis ($\delta = 0.95$). It is designed for low-risk, human-in-the-loop environments.

Configuration Parameters:

- ▶ **Trust Decay (δ):** 0.95. Maintained $>50\%$ trust retention even after 13 delegation hops.
- ▶ **Firewall Sensitivity:** Relaxed ($\tau = 0.9$).
- ▶ **Consensus:** Simple Majority.

Observed Performance (Part 2, Parameter Sensitivity Analysis):

- ▶ **Latency Overhead:** Minimal ($\sim 15\%$ baseline).
- ▶ **Detection Rate:** **87%** (vs 94% optimal).
- ▶ **Trade-off:** The high trust decay allows for fluid, deep delegation chains but increases vulnerability to subtle trust laundering (Ω_4).

Profile B: The “Customer Facing” Baseline (Balanced) I

This profile corresponds to the **Empirically Optimal Configuration** identified in Part 2 (Architecture-Specific Results), which balances security guarantees with operational overhead.

Configuration Parameters:

- ▶ **Trust Decay (δ):** 0.80. At this setting, trust degrades to <50% after 3 hops, strictly bounding the “radius of effective delegation.”
- ▶ **Firewall Sensitivity:** Balanced ($\tau = 0.5$).
- ▶ **Consensus:** Variable (Architecture Dependent).

Observed Performance (Part 2, Parameter Sensitivity Analysis):

- ▶ **Latency Overhead:** Reduced detection latency (~ 8.5 s for drift detection).
- ▶ **Detection Rate:** 94%.

Profile B: The “Customer Facing” Baseline (Balanced) II

- ▶ **Resilience:** Maximizes the F1 score, providing the best empirically observed trade-off between False Positives (0.06) and True Positives.

Profile C: The “Autonomous Operator” Baseline (High Assurance) I

This profile corresponds to the “Byzantine-Heavy” configuration tested in Part 2 (Byzantine Consensus Analysis). It is required for high-stakes, unsupervised environments.

Configuration Parameters:

- ▶ **Trust Decay (δ):** 0.60. Aggressive decay. Trust halves every 1.3 hops, enforcing a strictly flat command structure.
- ▶ **Firewall Sensitivity:** Strict ($\tau = 0.4$).
- ▶ **Consensus:** Byzantine Fault Tolerance ($n \geq 3f + 1$).

Observed Performance (Part 2, Parameter Sensitivity Analysis):

- ▶ **Latency Overhead:** Significant (+35-48% depending on corroboration count).
- ▶ **Detection Rate:** **95%** (general) to **98%** (LangGraph-specific).

Profile C: The “Autonomous Operator” Baseline (High Assurance) II

- ▶ **Resilience:** Specifically tunes parameters to resist coordination attacks (Ω_4), accepting higher latency for a near-zero semantic drift rate.

Architecture-Specific Observations I

Beyond the parameter profiles, Part 2's architecture adapters revealed specific interactions between the defense framework and the underlying agent topology.

LangGraph (State Machines)

Observation: LangGraph architectures achieved the highest overall detection rates (98%) in our tests (Part 2, Architecture-Specific Results). **Mechanism:** The explicit definition of state transitions allowed for rigorous **Invariant Checking**. Invalid state transitions were detected deterministically by the framework.

Architecture-Specific Observations II

CrewAI (Role-Based)

Observation: CrewAI architectures performed best against “Trust Exploitation” attacks (94% detection) (Part 2, Architecture-Specific Results). **Mechanism:** The framework’s role definitions acted as implicit **Identity Tripwires**. When an agent attempted to act outside its defined role, the behavior was flagged as a role violation.

Minimal Viable Implementation I

We also evaluated a “Minimal Viable Implementation” (MVI) to determine the baseline efficacy of the framework’s core components.

The Setup:

1. **Trust Decay:** $\delta = 0.80$ (The optimal balance point).
2. **Cognitive Firewall:** Ingress only.
3. **Tripwires:** One per agent.

Result: Even this minimal setup shifted the success rate against low-effort attacks from 100% (Baseline) to <5%, providing a critical first line of defense.

Alignment with Emerging Standards I

Practitioners deploying cognitive security must increasingly demonstrate compliance with industry and government standards. CIF's defense mechanisms map directly to two major 2025–2026 standardization efforts.

Alignment with Emerging Standards II

OWASP Top 10 for Agentic Applications (2026)

The OWASP Agentic Top 10 identifies ten risks (ASI01–ASI10) specific to autonomous multiagent deployments. CIF addresses the majority through its layered defense architecture:

OWASP Risk	CIF Defense	Profile Coverage
ASI01: Agent Goal Hijack	Cognitive Firewall + Tripwires	All profiles
ASI02: Tool Misuse/Exploitation	Belief Sandbox (tool output isolation)	Profiles B, C
ASI03: Identity/Privilege Abuse	Trust Calculus (δ^d decay)	All profiles
ASI06: Memory/Context Poisoning	Tripwire monitoring + Drift detection	Profiles B, C
ASI07: Insecure Inter-Agent Comm.	Provenance attestation	Profiles B, C